Myleran

Busulfan

QUALITATIVE AND QUANTITATIVE COMPOSITION

MYLERAN tablets containing busulfan 2mg.

- 2mg tablets are white, film-coated, round, biconvex tablets engraved “GX EF3” on one side and “M” on the other.

PHARMACEUTICAL FORM

Tablets

CLINICAL INFORMATION

Indications

Busulfan is indicated for the palliative treatment of the chronic phase of chronic myeloid leukaemia.

Busulfan is effective in producing prolonged remission in polycythaemia vera, particularly in cases with marked thrombocytosis.

Busulfan may be useful in selected cases of essential thrombocythaemia and myelofibrosis.

Dosage and Administration

Busulfan tablets are usually given in courses or administered continuously. The dose must be adjusted for the individual patient under close clinical and haematological control. Should a patient require an average daily dose of less than the content of the available busulfan tablets, this can be achieved by introducing one or more busulfan free days between treatment days. The tablets should not be divided (*see Instructions for Use/Handling*).

- **Obese Patients**

Dosing based on body surface area or adjusted ideal body weight should be considered in the obese (*see Pharmacokinetics*).

The relevant literature should be consulted for full details of treatment schedules.

CHRONIC MYELOID LEUKAEMIA

Populations

- **Induction in Adults**

Treatment is usually initiated as soon as the condition is diagnosed. The dose is 0.06 mg/kg/day, with an initial daily maximum of 4 mg, which may be given as a single dose.

There is individual variation in the response to busulfan and in a small proportion of patients the bone marrow may be extremely sensitive (*see Warnings and Precautions*).

The blood count must be monitored at least weekly during the induction phase and it may be helpful to plot counts on semilog graph paper.

The dose should be increased only if the response is inadequate after three weeks.

Treatment should be continued until the total leukocyte count has fallen to between 15 and 25 x 10^9 /l (typically 12 to 20 weeks). Treatment may then be interrupted, following which a further fall in the leukocyte count may occur over the next two weeks.

Continued treatment at the induction dose after this point or following depression of the platelet count to below 100 x 10^9 /l is associated with a significant risk of prolonged and possibly irreversible bone marrow aplasia.

- **Maintenance in Adults**

Control of the leukaemia may be achieved for long periods without further busulfan treatment; further courses are usually given when the leukocyte count rises to 50 x 10^9 /l, or symptoms return.

Some clinicians prefer to give continuous maintenance therapy. Continuous treatment is more practical when the duration of unmaintained remissions is short.

The aim is to maintain a leukocyte count of 10 to 15 x 10^9 /l and blood counts must be performed at least every four weeks. The usual maintenance dosage is on average 0.5 to 2 mg/day, but individual requirements may be much less. Should a patient require an average daily dose of less than the content of one tablet, the maintenance dose may be adjusted by introducing one or more busulfan free days between treatment days.

NOTE: lower doses of busulfan should be used if it is administered in conjunction with other cytotoxic agents (*see Interactions and Adverse Reactions*).

- **Paediatric Population**

Chronic myeloid leukaemia is rare in the paediatric age group.

Busulfan may be used to treat Philadelphia chromosome positive (Ph' positive) disease, but the Ph' negative juvenile variant responds poorly.

POLYCYTHAEMIA VERA

The usual dosage is 4 to 6 mg daily, continued for four to six weeks, with careful monitoring of the blood count, particularly the platelet count.

Further courses are given when relapse occurs; alternatively, maintenance therapy may be given using approximately half the induction dose.

If the polycythaemia is controlled primarily by venesection, short courses of busulfan may be given solely to control the platelet count.

MYELOFIBROSIS
The usual initial dose is 2 to 4 mg daily.

Very careful haematological control is required because of the extreme sensitivity of the bone marrow in this condition.

ESSENTIAL THROMBOCYTHAEMIA
The usual dosage is 2 to 4 mg per day.

Treatment should be interrupted if the total leukocyte count falls below 5 x 10^9 /l or the platelet count below 500 x 10^9 /l.

Contraindications
Busulfan should not be used in patients whose disease has demonstrated resistance to busulfan.

Busulfan should not be given to patients who have previously suffered a hypersensitivity reaction to busulfan or any other component of the preparation.

Warnings and Precautions
BUSULFAN IS AN ACTIVE CYTOTOXIC AGENT FOR USE ONLY UNDER THE DIRECTION OF PHYSICIANS EXPERIENCED IN THE ADMINISTRATION OF SUCH AGENTS.

Immunisation using a live organism vaccine has the potential to cause infection in immunocompromised hosts. Therefore, immunisations with live organism vaccines are not recommended.

Busulfan should be discontinued if lung toxicity develops (*see Adverse Reactions*).

Busulfan should not generally be given in conjunction with or soon after radiotherapy.

Busulfan is ineffective once blast transformation has occurred.

If anaesthesia is required in patients with possible pulmonary toxicity, the concentration of inspired oxygen should be kept as low as safely possible and careful attention given to post-operative respiratory care.

Hyperuricaemia and/or hyperuricosuria are not uncommon in patients with chronic myeloid leukaemia and should be corrected before starting treatment with busulfan. During treatment, hyperuricaemia and the risk of uric acid nephropathy should be prevented by adequate prophylaxis, including adequate hydration and the use of allopurinol.

Studies in renally impaired patients have not been conducted, however, as busulfan is moderately excreted in the urine, dose modification is not recommended in these patients. However, caution is recommended.

Busulfan has not been studied in patients with hepatic impairment. Since busulfan is mainly metabolised through the liver, caution should be observed when busulfan is used in patients with pre-existing impairment of liver function, especially in those with severe hepatic impairment.

The medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Conventional dose Treatment

Patients co-administered itraconazole or metronidazole with conventional dose busulfan should be monitored closely for signs of busulfan toxicity. Weekly measurements of blood counts are recommended when co-administering these drugs (*see Interactions*).

High-dose Treatment

If high-dose busulfan is prescribed, patients should be given prophylactic anticonvulsant therapy with preferably a benzodiazepine rather than phenytoin (*see Adverse Reactions and Interactions*).

Concomitant administration of itraconazole or metronidazole with high-dose busulfan has been reported to be associated with an increased risk of busulfan toxicity (*see Interactions*). Co-administration of metronidazole and high dose busulfan is not recommended. Co-administration of itraconazole with high dose busulfan should be at the discretion of the prescribing physician and should be based on a risk/benefit assessment.

Hepatic veno-occlusive disease is a major complication that can occur during treatment with busulfan. Patients who have received prior radiation therapy, greater than or equal to three cycles of chemotherapy, or prior progenitor cell transplant may be at an increased risk. (*see Adverse Reactions*).

A reduced incidence of hepatic veno-occlusive disease and other regimen-related toxicities have been observed in patients treated with busulfan and cyclophosphamide when the first dose of cyclophosphamide has been delayed for more than 24 h after the last dose of busulfan.

Monitoring
Careful attention must be paid to monitoring the blood counts throughout treatment to avoid the possibility of excessive myelosuppression and the risk of irreversible bone marrow aplasia (*see Adverse Reactions*).

Safe Handling of Busulfan Tablets
See Instructions for Use/Handling.

Mutagenicity
Various chromosome aberrations have been noted in cells from patients receiving busulfan.

Carcinogenicity
On the basis of short-term tests, busulfan has been classified as potentially carcinogenic by the International Agency for Research on Cancer (IARC). The World Health Association has concluded that there is a causal relationship between busulfan exposure and cancer.

Widespread epithelial dysplasia has been observed in patients treated with long-term busulfan, with some of the changes resembling pre-cancerous lesions.

A number of malignant tumours have been reported in patients who have received busulfan treatment.

The evidence is growing that busulfan, in common with other alkylating agents, is leukaemogenic. In a controlled prospective study in which two years' busulfan treatment was given as an adjuvant to surgery for lung cancer, long-term follow up showed an increased incidence of acute leukaemia compared with the placebo-treated group. The incidence of solid tumours was not increased.

Although acute leukaemia is probably part of the natural history of polycythaemia vera, prolonged alkylating agent therapy may increase the incidence.

Very careful consideration should be given to the use of busulfan for the treatment of polycythaemia vera and essential thrombocythaemia in view of the drug's carcinogenic potential. The use of busulfan for these indications should be avoided in younger or asymptomatic patients. If the drug is considered necessary, treatment courses should be kept as short as possible.

Oogenesis and spermatogenesis

Busulfan interacts with oogenesis and spermatogenesis. It may cause sterility in both sexes. Men treated with busulfan should be informed about sperm preservation prior to treatment (*see Pregnancy and Lactation; Fertility and Adverse Reactions*).

Interactions

Vaccinations with live organism vaccines are not recommended in immunocompromised individuals (*see Warnings and Precautions*).

The effects of other cytotoxics producing pulmonary toxicity may be additive (*see Adverse Reactions*).

The administration of phenytoin to patients receiving high-dose busulfan may result in a decrease in the myeloblative effect.

In patients receiving high-dose busulfan it has been reported that co-administration of itraconazole decreases clearance of busulfan by approximately 20% with corresponding increases in plasma busulfan levels.

Metronidazole has been reported to increase trough levels of busulfan by approximately 80%. Fluconazole had no effect on busulfan clearance. Consequently, high-dose busulfan in combination with itraconazole or metronidazole is reported to be associated with an increased risk of busulfan toxicity (*see Warnings and Precautions*).

A reduced incidence of hepatic veno-occlusive disease and other regimen-related toxicities have been observed in patients treated with high-dose busulfan and cyclophosphamide when the first dose of cyclophosphamide has been delayed for more than 24 h after the last dose of busulfan.

Paracetamol is described to decrease glutathione levels in blood and tissues, and may therefore decrease busulfan clearance when used in combination.

Increases in busulfan exposure have been observed at concomitant administration of busulfan and deferasirox. The mechanism behind the interaction is not fully elucidated. It is recommended to regularly monitor busulfan plasma concentrations and, if necessary, adjust the busulfan dose in patients who are or have recently been treated with deferasirox.

Pregnancy and Lactation

Fertility

Busulfan can lead to suppression of ovarian function and amenorrhoea in women and suppression of spermatogenesis in men. It may cause sterility in both sexes. In women busulfan may cause severe and persistent ovarian failure, including failure to achieve puberty after administration to young girls and pre-adolescents at high-dose. It may also cause male infertility, azoospermia and testicular atrophy in male patients receiving busulfan. *(See Non-Clinical Information – Reproductive Toxicology).*

Pregnancy

As with all cytotoxic chemotherapy, adequate contraceptive precautions should be advised when either partner is receiving busulfan.

The use of busulfan should be avoided whenever possible during pregnancy, particularly during the first trimester.

In every individual case the expected benefit of treatment to the mother must be weighed against the possible risks to the foetus.

A few cases of congenital abnormalities, not necessarily attributable to busulfan, have been reported and third trimester exposure may be associated with impaired intra-uterine growth. However, there have also been many reported cases of apparently normal children born after exposure to busulfan in utero, even during the first trimester.

Studies of busulfan treatment in animals have shown reproductive toxicity (*see Non-Clinical Information).* The potential risk for humans is largely unknown.

Lactation

It is not known whether busulfan or its metabolites are excreted in human breast milk. Mothers receiving busulfan should not breast feed their infants.

Ability to perform tasks that require judgement, motor or cognitive skills

There are no data on the effect of busulfan on driving performance or the ability to operate machinery. A detrimental effect on these activities cannot be predicted from the pharmacology of the drug.

Adverse Reactions

For this product there is no modern clinical documentation which can be used as support for determining the frequency of undesirable effects. Undesirable effects may vary in their incidence depending on the dose received and also when given in combination with other therapeutic agents.

The following convention has been utilised for the classification of frequency: Very common (≥1/10), common (≥1/100 and <1/10), uncommon (≥1/1000 and <1/100), rare (≥1/10,000 and <1/1000), very rare (<1/10,000) and, Not known (cannot be estimated from the available data).

The following table of adverse reactions originated from the use of busulfan, or busulfan in combination with other therapeutic agents.

System organ class	**Frequency**	**Side effects**
Neoplasms benign, malignant and unspecified (including cysts and polyps)	Common	Leukaemia secondary to oncology chemotherapy (*see Warnings and Precautions; Carcinogenicity*)
Blood and lymphatic system disorders	Very common	Dose-related bone marrow failure, manifesting as leukopenia and particularly thrombocytopenia
	Rare	Aplastic anaemia
Nervous system disorders	Rare	At high-dose: seizure (*see Warnings and Precautions* and *Interactions*)
	Very rare	Myasthenia gravis
Eye disorders	Rare	Lens disorder and cataract, (which may be bilateral) corneal thinning (reported after bone marrow transplantation preceded by high-dose busulfan treatment)
Cardiac disorders	Common	At high-dose: cardiac tamponade in patients with thalassaemia
Respiratory, thoracic and mediastinal disorders	Very common	At high-dose: idiopathic pneumonia syndrome
	Common	Interstitial lung disease following long term conventional dose use
Gastrointestinal disorders	Very common	At high-dose: nausea, vomiting, diarrhoea, mouth ulceration
	Rare	At conventional dose: nausea, vomiting, diarrhoea, mouth ulceration, which may possibly be ameliorated by using divided doses. Dry mouth
	Not known	Tooth hypoplasia
Hepatobiliary disorders	Very common	At high-dose: hyperbilirubinaemia, jaundice, veno-occlusive liver disease (*see Warnings and Precautions* and *Interactions*) and biliary fibrosis with hepatic atrophy and hepatic necrosis
	Rare	Jaundice and abnormal hepatic function, at conventional dose. Biliary fibrosis
Skin and subcutaneous tissue disorders	Common	Alopecia at high-dose. Skin hyperpigmentation (*see also General disorders and administration site conditions*)

	Rare	Alopecia at conventional dose, skin reactions including urticaria, erythema multiforme, erythema nodosum, porphyria non-acute, rash, dry skin and skin fragility with complete anhydrosis cheilosis
Musculoskeletal and connective tissue disorders	Rare	Sjögren's syndrome
Injury, poisoning and procedural complications	Rare	Radiation skin injury is increased in patients receiving radiotherapy soon after high-dose busulfan
Renal and urinary disorders	Common	At high-dose: cystitis haemorrhagic in combination with cyclophosphamide
Reproductive system and breast disorders *	Very common	Ovarian disorder and amenorrhoea with menopausal symptoms in pre-menopausal patients at high-dose; severe and persistent ovarian failure, including pubertal failure after administration to young girls and pre-adolescents at high-dose. Male infertility, azoospermia and testicular atrophy in male patients receiving busulfan
	Uncommon	Ovarian disorder and amenorrhoea with menopausal symptoms in pre-menopausal patients at conventional dose
	Very rare	Gynaecomastia
General disorders and administration site conditions	Rare	Dysplasia

Description of selected adverse events

Blood and lymphatic system disorders

Aplastic anaemia (sometimes irreversible) has been reported rarely, typically following long-term conventional doses and also high doses of busulfan.

Respiratory, thoracic and mediastinal disorders

Pulmonary toxicity after either high or conventional dose treatment typically presents with non-specific non-productive cough, dyspnoea and hypoxia with evidence of abnormal pulmonary physiology. Other cytotoxic agents may cause additive lung toxicity (*see Interactions*). It is possible that subsequent radiotherapy can augment subclinical lung injury caused by busulfan. Once pulmonary toxicity is established the prognosis is poor despite busulfan withdrawal and there is little evidence that corticosteroids are helpful.

Idiopathic pneumonia syndrome is a non-infectious diffuse pneumonia which usually occurs within three months of high dose busulfan conditioning prior to allogeneic or autologous haemopoietic transplant. Diffuse alveolar haemorrhage may also be detected in some cases after broncholavage. Chest X-rays or CT scans show diffuse or non-specific focal infiltrates and biopsy shows interstitial pneumonitis and diffuse alveolar damage and sometimes fibrosis.

Interstitial pneumonitis may occur following conventional dose use and lead to pulmonary fibrosis. This usually occurs after prolonged treatment over a number of years. The onset is usually insidious but may also be acute. Histological features include atypical changes of the alveolar and bronchiolar epithelium and the presence of giant cells with large hyperchromatic nuclei. The lung pathology may be complicated by superimposed infections. Pulmonary ossification and dystrophic calcification have also been reported.

<u>Hepatobiliary disorders</u>
Busulfan is not generally considered to be significantly hepatotoxic at normal therapeutic doses. However, retrospective review of post mortem reports of patients who had been treated with low-dose busulfan for at least two years for chronic myeloid leukaemia showed evidence of centrilobular sinusoidal fibrosis.

<u>Skin and subcutaneous tissue disorders</u>
Hyperpigmentation occurs, particularly in those with a dark complexion. It is often most marked on the neck, upper trunk, nipples, abdomen and palmar creases. This may also occur as part of a clinical syndrome (*see General disorders and administration site conditions*).

<u>Reproductive system and breast disorders</u>
Studies of busulfan treatment in animals have shown reproductive toxicity (*see Non-Clinical Information*)

<u>General disorders and administration site conditions</u>
Clinical syndrome (weakness, severe fatigue, anorexia, weight loss, nausea and vomiting and hyperpigmentation of the skin) resembling adrenal insufficiency (Addison’s disease) but without biochemical evidence of adrenal suppression, mucous membrane hyperpigmentation or alopecia (*see Skin and subcutaneous tissue disorders*) has been seen in a few cases following prolonged busulfan therapy. The syndrome has sometimes resolved when busulfan has been withdrawn.

Many histological and cytological changes have been observed in patients treated with busulfan, including widespread dysplasia affecting uterine cervical, bronchial and other epithelia. Most reports relate to long-term treatment but transient epithelial abnormalities have been observed following short-term, high-dose treatment.

Overdosage

Symptoms and Signs

The acute dose-limiting toxicity of busulfan in man is myelosuppression.

The main effect of chronic overdosage is bone marrow depression and pancytopenia.

Treatment

There is no known antidote. Dialysis should be considered in the management of overdose as there is one report of successful dialysis of busulfan.

Appropriate supportive treatment should be given during the period of haematological toxicity.

Clinical Pharmacology

Pharmacodynamics

ATC Code

Pharmaceutical group: Alkyl sulfonates; ATC code L01AB01

Mechanism of Action

Busulfan (1,4-butanediol dimethanesulfonate) is a bifunctional alkylating agent. Binding to DNA is believed to play a role in its mode of action, and di-guanyl derivatives have been isolated, but interstrand crosslinking has not been conclusively demonstrated.

The basis for the uniquely selective effect of busulfan on granulocytopoiesis is not fully understood. Although not curative, busulfan is very effective in reducing the total granulocyte mass, relieving the symptoms of disease and improving the clinical state of the patient. Busulfan has been shown to be superior to splenic irradiation when judged by survival times and maintenance of haemoglobin levels and is as effective in controlling spleen size.

Pharmacodynamic Effects

Not applicable.

Pharmacokinetics

Absorption

The area under the curve (AUC) and peak plasma concentrations (C_{max}) of busulfan have been shown to be linearly dose dependent. Following administration of a single 2 mg oral dose of busulfan, the AUC and C_{max} of busulfan were 125±17 nanograms.h/ml and 28±5 nanograms/ml respectively.

A lag time between busulfan administration and detection in the plasma of up to 2 h has been reported.

The bioavailability of oral busulfan shows large intra-individual variation ranging from 47% to 103% (mean 80%) in adults.

High-dose Treatment

Drug was assayed either using gas liquid chromatography with electron capture detection or by high-performance liquid chromatography (HPLC).

Following oral administration of high dose busulfan (1 mg/kg every 6 h for 4 days), AUC and C_{max} in adults are highly variable but have been reported to be 8260 nanograms.h/ml (range 2484 to 21090) and 1047 nanograms/ml (range 295 to 2558) respectively when measured by HPLC and 6135 nanograms.h/ml (range 3978 to 12304) and 1980 nanograms/ml (range 894 to 3800) respectively using gas chromatography.

Distribution

Busulfan is reported to have a volume of distribution of 0.64±0.12 L/kg in adults.

Busulfan given in high doses has been shown to enter the cerebrospinal fluid (CSF) in concentrations comparable to those found in plasma, with a mean CSF : plasma ratio of 1.3 : 1. The saliva : plasma distribution of busulfan was 1.1 : 1.

The level of busulfan bound reversibly to plasma proteins has been variably reported to be insignificant or approximately 55%. Irreversible binding of drug to blood cells and plasma proteins has been reported to be 47% and 32%, respectively.

Metabolism

Busulfan metabolism involves a reaction with glutathione, which occurs via the liver and is mediated by glutathione-S-transferase.

The urinary metabolites of busulfan have been identified as 3-hydroxysulpholane, tetrahydrothiophene 1-oxide and sulpholane, in patients treated with high-dose busulfan.

Elimination

Busulfan has a mean elimination half life of between 2.3 and 2.8 h. Adult patients have demonstrated a clearance of busulfan of 2.4 to 2.6 ml/min/kg. The elimination half life of busulfan has been reported to decrease upon repeat dosing suggesting that busulfan potentially increases its own metabolism.

Very little (1 to 2%) busulfan is excreted unchanged in the urine.

Special Patient Populations

Children

The bioavailability of oral busulfan shows large intra-individual variation ranging from 22% to 120% (mean 68%) in children.

Plasma clearance is reported to be 2 to 4 times higher in children than in adults when receiving 1 mg/kg every 6 h for 4 days. Dosing children according to body surface area has been shown to give AUC and C_{max} values similar to those seen in adults. The area under the curve has been shown to be half that of adults in children under the age of 15 years and a quarter of that of adults in children under 3 years of age.

Busulfan is reported to have a volume of distribution of 1.15±0.52 L/kg in children.

When busulfan is administered at a dose of 1 mg/kg every 6 h for 4 days, the CSF : plasma ratio has been shown to be 1.02 : 1. However, when administered at a dose of 37.5 mg/m^2 every 6 h for 4 days the ratio was 1.39 : 1.

Obese Patients

Obesity has been reported to increase busulfan clearance. Dosing based on body surface area or adjusted ideal bodyweight should be considered in the obese.

Clinical Studies

Not applicable.

NON-CLINICAL INFORMATION

- **Carcinogenesis, mutagenesis**

Busulfan has been shown to be mutagenic in various experimental systems, including bacteria (Ames Salmonella test), fungi, Drosophila and cultured mouse lymphoma cells.

In vivo cytogenetic studies in rodents have shown an increased incidence of chromosome aberrations in both germ cells and somatic cells after busulfan treatment.

There is insufficient evidence from preclinical studies to determine whether busulfan has carcinogenic potential *(see Warnings and Precautions*).

- **Reproductive toxicology**

There is evidence from animal studies that busulfan produces foetal abnormalities and adverse effects on offspring, including defects of the musculo-skeletal system, reduced body weight and size, impairment of gonadal development and effects on fertility.

Busulfan interferes with spermatogenesis in experimental animals. Limited studies in female animals indicate busulfan has a marked and irreversible effect on fertility via oocyte depletion.

PHARMACEUTICAL INFORMATION

List of Excipients

2 mg Tablets:	
Tablet core:	**Tablet coating:**
Anhydrous lactose Pregelatinised starch Magnesium stearate	Hypromellose Titanium dioxide Triacetin

Incompabilities

None known.

Shelf Life

The expiry date is indicated on the packaging.

Special Precautions for Storage

Store below 30ºC.

Nature and Contents of Container

Busulfan tablets are supplied in amber glass bottles (25's) with a child resistant closure.

Instructions for Use/Handling

Safe handling of busulfan tablets:

The tablets should not be divided and provided the outer coating is intact, there is no risk in handling busulfan tablets.

Handlers of busulfan tablets should follow guidelines for the handling of cytotoxic drugs according to prevailing local recommendations and/or regulations.

Disposal:

Busulfan tablets surplus to requirements should be destroyed in a manner appropriate to the prevailing local regulations for the destruction of dangerous substances.

Not all presentations are available in every country.

Version number of package insert: CCDS17/02/0125
Date of revision of package insert: 15 Jan 2025

aspen

MANUFACTURER

Excella GmbH & Co. KG, Feucht, Germany.